Spatiotemporal Point-Process Modeling of Domestic Violence in Minneapolis

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Introduction

Aggravated assaults are a significant public health concern. Understanding spatiotemporal patterns in domestic and non-domestic assaults is essential for targeted prevention and policy making.

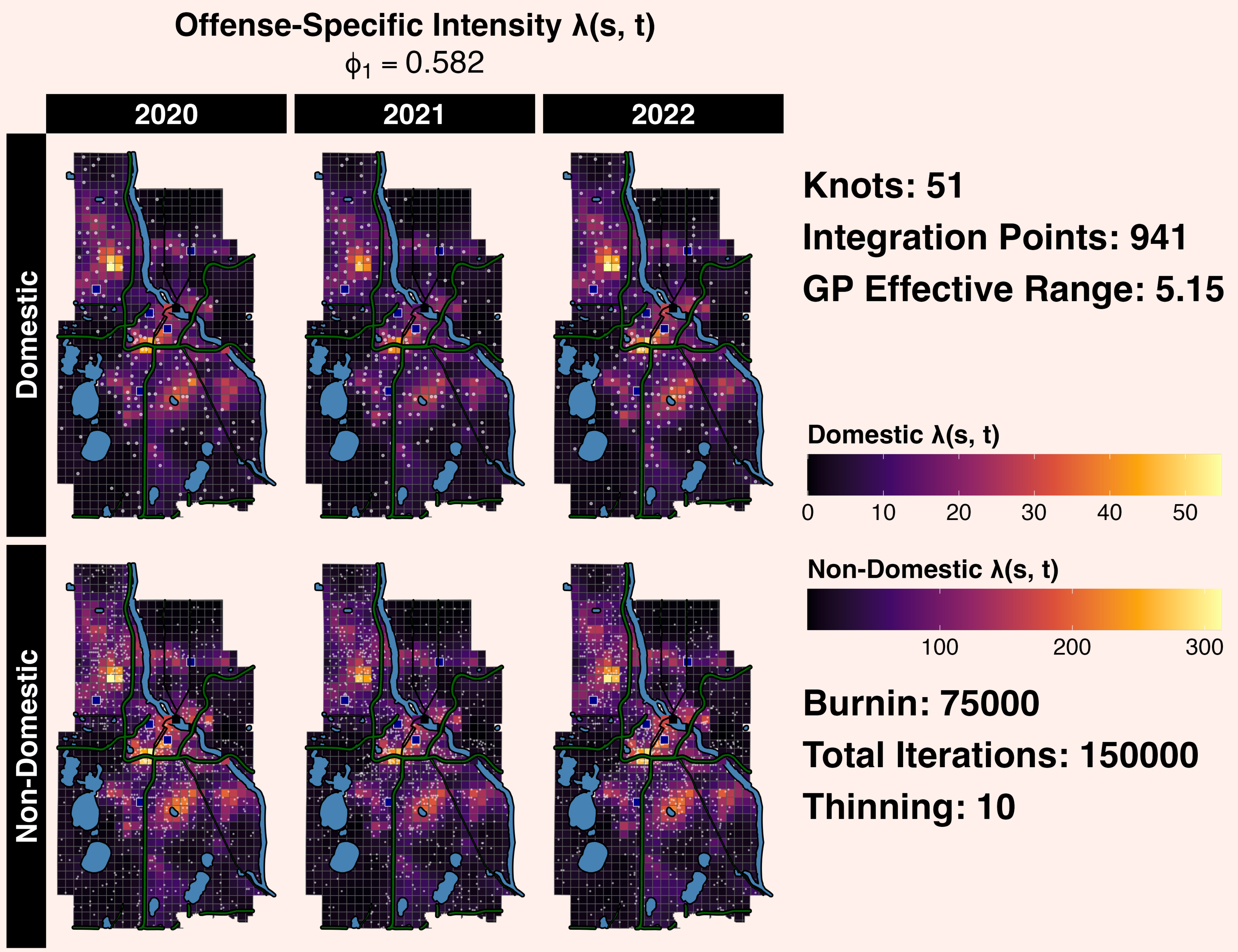
Objectives:

- Model spatial and temporal patterns of aggravated assaults (2020–2022) in Minneapolis.
- Estimate offense-specific intensities (Domestic vs Non-Domestic) while adjusting for spatial covariates (distance from bodies of water, % of 911 calls that sound domestic)

Study Design and Population:

- All aggravated assaults taking place in Minneapolis captured in police reports (2020–2022)
- Derived spatial covariates (distance from water) and percent of calls that may be flagged as domestic (not equivalent to actual domestic incidents)
- JSD of 0.1 $\Longrightarrow$ moderately strong similarity between domestic/non-domestic patterns

Posterior Surfaces



Methods

Model Overview: Modeled the location/intensity of aggravated assaults as well as $\pi(s_i, t_i)$, the probability of those assaults being domestic assaults while accounting for yearly changes. For the intensity, a GP with an exponential kernel and $n_K = 51$ basis functions, was used to capture spatial patterns.

Window: $\mathcal{V} = \mathcal{D} \times \mathcal{T} = \{\text{land area of MPLS}\} \times \{2020, \dots, 2022\}$

$$\lambda_{\mathcal{D}\times\mathcal{T}} = \int_{\mathcal{T}}\int_{\mathcal{D}} \lambda(\boldsymbol{s}, t)\,d\boldsymbol{s}\,dt = \int_{\mathcal{V}} \lambda(\boldsymbol{v})\,d\boldsymbol{v} \approx \sum_{i=1}^{N} a_i \lambda(\boldsymbol{v}_i)$$

LGCP Intensity Model:

$$\{(\mathbf{s}_i, t_i)\}_{i=1}^{N} \sim \mathrm{LGCP}(\lambda_{\mathcal{D}\times\mathcal{T}})$$
$$\log\{E(\mathbf{s}_i)\lambda(\mathbf{s}_i, t_i)\} = \mathbf{x}_1(\mathbf{s}_i)^\top \boldsymbol{\beta} + \boldsymbol{w}(\mathbf{s}_i) + \delta_1(t_i)$$
$$\boldsymbol{\beta} \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, 1.33^2\mathbf{I}),$$
$$\boldsymbol{\delta}_1|\tau_1^2 = \tau_1 \cdot \boldsymbol{u}_1 \sim \mathcal{N}(\mathbf{0}, \tau_1^2 \boldsymbol{I}), \quad \mathbf{1}^T \boldsymbol{u}_1 = 0$$
$$\log \tau_1 \sim \mathcal{N}\left(log(\hat{\tau}_1), 0.22^2\right)$$

Mark (Offense Type) Model:

$$Y_i|\pi(s_i, t_i) \sim \mathrm{Bern}(\pi(s_i, t_i))$$
$$\mathrm{logit}\ \pi(s_i, t_i) = \mathbf{x}_2(\mathbf{s}_i, t_i)^\top \boldsymbol{\gamma} + \delta_2(t_i)$$
$$\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, 1.21^2\mathbf{I}),$$
$$\boldsymbol{\delta}_2|\tau_2^2 = \tau_2 \cdot \boldsymbol{u}_2 \sim \mathcal{N}(\mathbf{0}, \tau_2^2 \boldsymbol{I}), \quad \mathbf{1}^T \boldsymbol{u}_2 = 0$$
$$\log \tau_2 \sim \mathcal{N}\left(log(\hat{\tau}_2), 0.22^2\right)$$

$\pi(s_i, t_i)$ thins the overall aggravated assault intensity surface $\lambda(s_i, t_i)$ to derive offense-specific intensity surfaces $\lambda_1(s_i, t_i) = \pi(s_i, t_i) \cdot \lambda(s_i, t_i)$ and $\lambda_0(s_i, t_i) = [1 - \pi(s_i, t_i)] \cdot \lambda(s_i, t_i)$.

Posterior Summaries

	Intensity Model (Rel. Risk)		Mark Model (Odds Ratio)		
Parameter	Mean	95% CI	Mean	95% CI	Parameter
Intercept	41.253	(40.484, 42.028)	0.150	(0.144, 0.157)	Intercept
Distance from Water	0.457	(0.443, 0.472)	1.076	(1.035, 1.118)	Distance from Water
	—	——	1.114	(1.071, 1.158)	"Domestic" Call
Year 2020	1.053	(1.035, 1.071)	0.964	(0.922, 1.007)	Year 2020
Year 2021	0.928	(0.911, 0.945)	0.987	(0.945, 1.031)	Year 2021
Year 2022	1.024	(1.006, 1.042)	1.050	(1.005, 1.100)	Year 2022
Spatial SD	2.212	(1.920, 2.525)	—-	——	Spatial SD
Year SD	0.059	(0.039, 0.086)	0.047	(0.030, 0.069)	Year SD

Conclusions

- Joint LGCP + Logit model captures space-time variation (year effects indicate gradual change).
- Assaults in general are more prevalent near areas of water (likely due to Downtown being close to the river)
- Domestic offenses linked to areas further away from water and areas with a higher percentage domestic-flagged calls
- **Limitations**: No GP for mark model; β estimates quite different from frequentist GLM
- **Future work**: finer covariates and alternative spatial priors

References

- City of Minneapolis. Crime Data. `https://opendata.minneapolismn.gov/datasets/cityoflakes::crime-data/about`
- Kelling & Haran. (2022). A two-stage Cox process model with spatial and nonspatial covariates. *JRSS B*.